

Predictive Analytics Using Historical Data

1. Project

Predictive Analytics Using Historical Data Using Linear Regression

2. Objective

The main goal of this project is to create a model that can predict what sales will be like in the future. We are using sales data to make this model. The project uses Linear Regression to figure out what sales trends will be like in the future. This helps businesses make decisions based on what happened in the past.

3. Dataset Description

The dataset has daily sales records with the following information:

- Date. The day each sale was made
- Sales. The sales for that day

We made the dataset in Microsoft Excel and then moved it to Python to look at the data and make predictions.

4. Data Preprocessing

We did the following steps to get the data ready for the model:

- We brought the dataset into Python using Pandas.
- We changed the Date column to a format that computers can understand.
- We turned the dates into numbers that the computer can use for machine learning.
- We checked the dataset to make sure it was consistent and ready for the model.

5. Tools & Technologies Used

- Python
- Pandas
- Scikit-learn
- Matplotlib
- Microsoft Excel
- Google Colab / Visual Studio Code

6. Machine Learning Model Used

We used Linear Regression for this project. Linear Regression is a type of machine learning that helps predict numbers. In this project we used it to find the relationship between dates and sales numbers so we can predict future sales.

7. Model Evaluation

We used the following to see how well the model worked:

- Mean Squared Error. This measures how our predictions are from the real numbers.
- R^2 Score. This measures how well the model explains the changes in sales data.

These numbers help us see how accurate our predictions are.

8. Visualizations Used

The project includes the following pictures:

- A line chart that shows the sales trend
- A chart that compares the sales numbers to the predicted numbers

These pictures help us compare old sales to predicted sales and see what the sales trends are.

9. Business Insights

The sales data shows that sales have been going up slowly over time.

The Linear Regression model did a job of finding the relationship between the date and sales.

The predicted sales numbers are close, to the sales numbers.

Businesses can use these predictions to plan their inventory, budget and future sales.

Using analytics helps companies make decisions based on data.

10. Conclusion

This project shows that we can use analytics on old sales data. By using the Linear Regression algorithm we can estimate sales numbers with good accuracy. The project highlights the importance of getting the data ready evaluating the model and making pictures to show the data. The predictions we made can help businesses plan better and make decisions.

11. Outputs:

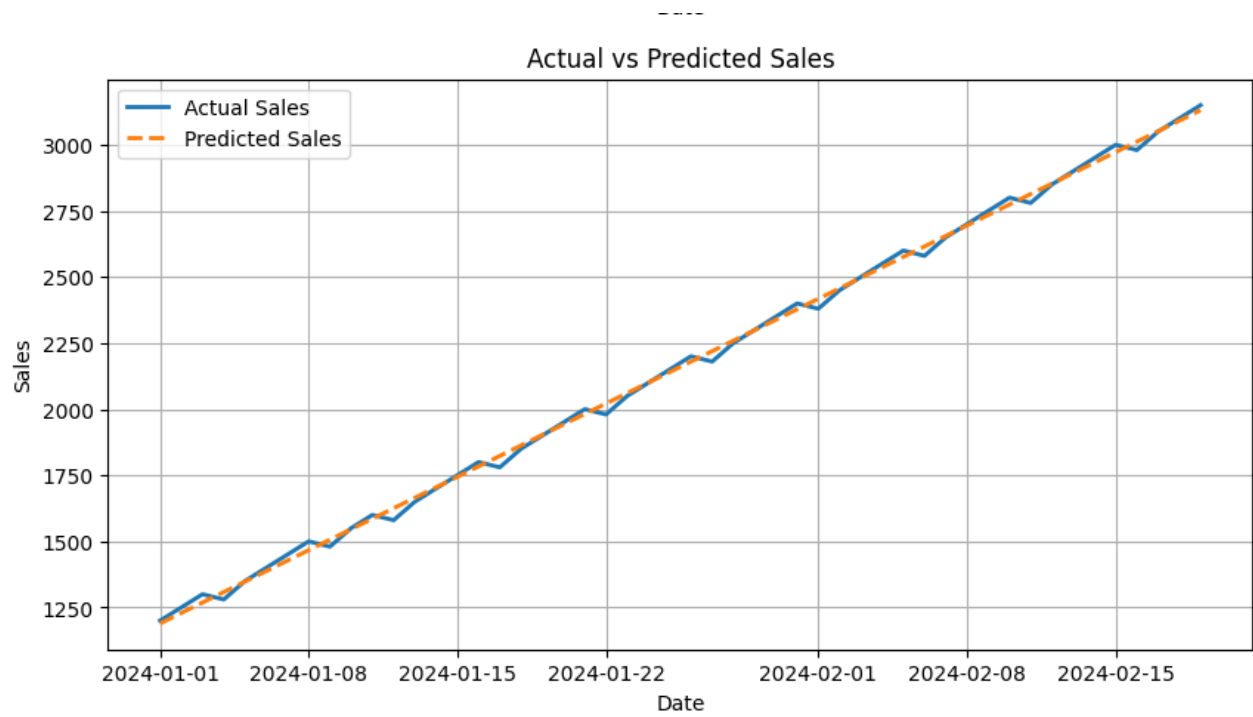
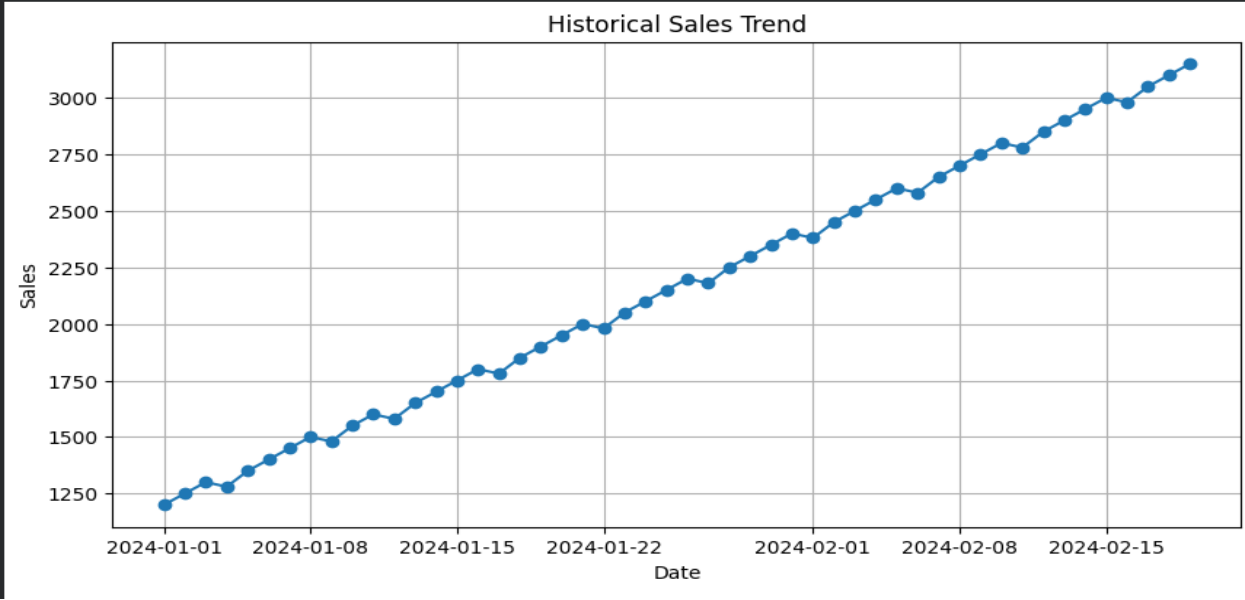
Dataset Preview:

	Date	Sales
0	2024-01-01	1200
1	2024-01-02	1250
2	2024-01-03	1300
3	2024-01-04	1280
4	2024-01-05	1350

Model Evaluation

Mean Squared Error : 457.57

R2 Score : 0.9986



Future Sales Prediction

	Future_Date	Predicted_Sales
0	2024-02-20	3170.253061
1	2024-02-21	3209.870828
2	2024-02-22	3249.488595
3	2024-02-23	3289.106363
4	2024-02-24	3328.724130
5	2024-02-25	3368.341897
6	2024-02-26	3407.959664
7	2024-02-27	3447.577431
8	2024-02-28	3487.195198
9	2024-02-29	3526.812965

Project Completed Successfully!

Generated Files:

1. Historical_Sales_Predictions.xlsx
2. Future_Sales_Prediction.xlsx
3. sales_trend.png
4. actual_vs_predicted.png